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EDUCATION

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science

Expected June 2028

Los Angeles, CA

EXPERIENCE

Software Engineering Intern (Incoming)

Google (YouTube)

June 2026 – Sept 2026

San Bruno, CA

Software Engineering Intern

VideoLAN

June 2025 – Sept 2025

Remote

- Developed C/C++ video filters (real-time visual effects and processing) for VLC Media Player (traffic cone).
- Integrated a SAM2 model module with OpenCV to enable real-time object segmentation for visual analysis.
- Implemented YuNet-based ML face detection for real-time tracking of 100+ faces in surveillance systems.
- Built fast video-to-GIF conversion with 5 dithering algorithms for customizable quality and performance.
- Analyzed the AV1 video-codec specification and studied dav1d decoding algorithms to deepen knowledge.

Software Developer Intern

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Feb. 2023 – June 2023

Remote

- Designed MongoDB-based TypeScript RESTful APIs to fetch customer transactions and emails.
- Launched a JavaScript Discord bot to link 100+ customers' WHMCS accounts with Discord.
- Established secure OAuth2 gateway RESTful API using Express.js for account verification.

Java Mentor

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Sept. 2021 – Feb. 2022

Remote

- Taught 20 weekly 1-hour lessons to 10 middle/high-school students with varying coding experience.
- Raised nearly \$1,000 dollars for the non-profit, helping the school keep afloat during COVID-19.
- Spent 1-2 additional hours weekly preparing homework/slides to help students master material.

PROJECTS

XXXXXXXXXXXXXX | *C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression*

Feb. 2026 – Present

- Popular C++ tool (870+ stars, 110 forks) that encodes local files into YouTube videos for storage.
- Achieved a 100x speed-up using inline Assembly, SIMD intrinsics, and OpenMP parallelization.
- Designed lightweight Qt 6 UI and libsodium encryption, enhancing user experience and security.
- Uses robust CRC/Wirehair error-correction to novelly bypass YouTube's compression algorithm.
- Reached #5 on Hacker News (Y Combinator) within 12 hours, generating thousands of views.
- Includes technical explanation video with over 3.3M impressions, educating over 270k viewers.

XXXX | *Java, Spring Boot, TypeScript, CI/CD*

June 2020 – Present

- Popular Java framework (130+ stars, 10 forks) for building fast and portable media applications.
- Integrates with VLC/mpv with a documented image-processing pipeline API for easier adoption.
- Uses Spring Boot to stream sub-100ms latency audio to usable TypeScript/Howler.js frontend.
- Configured CI/CD pipeline with TeamCity using Oracle Cloud to automate software releases.
- Engineered native media player via JavaCV that is 5x faster than most pure-Java players.

XXXXXXXXXXXXXX | *C++, OpenGL, OpenAL, FFmpeg*

Feb. 2026 – Present

- Robust C++ media player written in less than 1K lines of code with small UI.
- Built a GPU-accelerated OpenGL shader pipeline for smooth 1080p playback.
- Supports OpenAL for automated audio device selection and smooth playback.

TECHNICAL SKILLS

Languages: Java, C/C++, JavaScript/TypeScript, Assembly, OCaml

Frameworks & Libraries: Spring, Hibernate, FFmpeg, OpenCV, VLC, React, Next.js, Tailwind

Technologies: Git, MongoDB, Google Cloud, Oracle Cloud

Awards: USACO Gold, Google Foobar